

BIO 210

Cellular Biology

Structure, energy, information, and experimental reasoning at the cellular scale

FALL 2026 | AUGUST 24 - DECEMBER 18

Instructor	Mei Chen, PhD
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Class meetings	Lecture: Mon/Wed/Fri, 9:00-9:50 AM; Lab: Wed, 1:00-3:50 PM
Location	Science Center 105; Laboratory SC 142
Office hours	Mon 2:00-3:00 PM; Wed 10:00-11:30 AM; Fri by appointment
Credits	4 semester credits with laboratory
Prerequisites	BIO 110 and CHEM 120 with grades of C or better

Course description

BIO 210 investigates how cells maintain boundaries, transform energy, store and express information, communicate, divide, and respond to their environments. Lecture and laboratory are integrated: students use microscopy, quantitative assays, and evidence-based models to explain cellular behavior. Emphasis is placed on safe laboratory practice, transparent data handling, and the limits of experimental conclusions.

Learning outcomes

- Relate cellular structures to transport, signaling, energy conversion, and division.
- Predict how molecular or environmental changes affect cellular processes.
- Design controls and interpret quantitative data from cell-biology experiments.
- Maintain a reproducible laboratory notebook and communicate results in scientific form.
- Apply laboratory safety, waste-disposal, and research-integrity standards consistently.

This syllabus is a course roadmap. Dates may shift if campus operations or class learning needs require a change; any revision will be announced in class and in the learning portal.

Materials and learning tools

- Alberts et al., *Essential Cell Biology*, 6th edition. Earlier editions may be used with instructor approval.
- BIO 210 Laboratory Manual, Fall 2026, supplied digitally through the learning portal.
- Bound quadrille laboratory notebook, permanent black or blue ink, splash goggles, closed-toe shoes, and a knee-length lab coat.
- Basic calculator and spreadsheet software. Specialized analysis software is available in the Science Center computer lab.

How your grade is calculated

Category	Weight / points	What it measures
Problem Sets (8)	200 pts	Quantitative practice and mechanism-based predictions; 25 points each.
Laboratory Reports (3)	300 pts	Individual scientific reports using class or team data; 100 points each.
Unit Examinations (2)	250 pts	Two in-class exams; 125 points each.
Cumulative Final Examination	150 pts	Mechanism, data interpretation, and experimental design.
Laboratory Notebook Checks (3)	100 pts	Contemporaneous records, calculations, figures, and safety documentation.

Grade scale

A = 930-1000 | A- = 900-929 | B+ = 870-899 | B = 830-869 | B- = 800-829 | C+ = 770-799 | C = 730-769 | C- = 700-729 | D = 600-699 | F = below 600

Major assignment expectations

Problem sets

Eight problem sets combine short calculations, pathway diagrams, and explanations of experimental results. Each set includes a 'prediction before calculation' prompt to make reasoning visible. Work may be discussed with peers, but submitted explanations and figures must be individual.

Laboratory reports

Reports follow a concise scientific format: title, question, methods summary, results with a labeled figure or table, interpretation, and limitations. Reports address microscopy and measurement, enzyme kinetics, and membrane transport. Raw data and analysis files must be retained.

Unit and final examinations

Exams are completed in person on Wednesdays. Questions emphasize application rather than memorized definitions. One handwritten 4 x 6 inch reference card is allowed; cards are submitted with the exam.

Laboratory notebook

Entries must be made during the laboratory session and include dates, objectives, reagent calculations, procedural changes, raw observations, and initial interpretations. Neatness is not graded; completeness, traceability, and honesty are.

Assignment calendar

All dates below are firm unless the instructor announces a change. Submission times use Pacific Time. The recurring due-day pattern is shown explicitly so students can plan ahead.

Assignment	Type	Due	Pts
Problem Set 1: Scale, Units, and Microscopy	Problem set	Sun, Aug 30 11:59 PM	25
Problem Set 2: Macromolecular Interactions	Problem set	Sun, Sep 13 11:59 PM	25
Laboratory Report 1: Microscopy and Measurement	Lab report	Wed, Sep 23 11:59 PM	100
Problem Set 3: Membrane Transport	Problem set	Sun, Sep 27 11:59 PM	25
Laboratory Notebook Check 1	Notebook	Wed, Sep 30 3:50 PM	30
Unit Examination 1	Exam	Wed, Oct 7 9:00 AM	125
Problem Set 4: Enzyme Kinetics	Problem set	Sun, Oct 18 11:59 PM	25
Laboratory Report 2: Enzyme Kinetics	Lab report	Wed, Oct 28 11:59 PM	100
Problem Set 5: Cellular Respiration	Problem set	Sun, Nov 1 11:59 PM	25
Laboratory Notebook Check 2	Notebook	Wed, Nov 4 3:50 PM	30
Problem Set 6: Photosynthesis	Problem set	Sun, Nov 15 11:59 PM	25
Unit Examination 2	Exam	Wed, Nov 18 9:00 AM	125
Problem Set 7: Gene Expression	Problem set	Sun, Nov 29 11:59 PM	25
Laboratory Notebook Check 3	Notebook	Wed, Dec 2 3:50 PM	40
Problem Set 8: Cell Cycle and Signaling	Problem set	Sun, Dec 6 11:59 PM	25
Laboratory Report 3: Membrane Transport	Lab report	Wed, Dec 9 11:59 PM	100
Cumulative Final Examination	Final exam	Wed, Dec 16 8:00 AM	150

Submission checklist

Before submitting: confirm the correct file opened, include your name and course code, cite borrowed ideas, and retain a local copy. A portal timestamp is the official submission time.

Course policies and support

Laboratory safety and attendance

Students must complete the safety orientation before handling materials. Goggles, lab coats, closed-toe shoes, and appropriate leg coverage are required. Food, drinks, and headphones are prohibited in the lab. A student who misses a laboratory should notify the instructor before the session when possible; one alternate-data assignment is available, but experiments cannot generally be recreated individually.

Due-day rhythm and late work

Problem sets are due Sundays at 11:59 PM. Reports, notebook checks, and examinations occur on Wednesdays. Problem sets may be submitted up to 48 hours late for a maximum 15% deduction. Reports lose 10% per day for three days. Safety documentation and exams require an approved make-up arrangement.

Data integrity

Never erase, overwrite, fabricate, or silently discard observations. Corrections in paper notebooks use a single line with initials and date. Electronic files must preserve raw data in a separate sheet or file. Unexpected data are scientifically useful when reported accurately and discussed carefully.

Generative AI and computational tools

Generative AI may explain a general concept or help debug spreadsheet syntax, but it may not create report text, invent citations, produce data, or interpret the class dataset. Any permitted use must be documented in an appendix containing the prompt, relevant output, and a statement of verification. Grammar and accessibility tools are permitted.

Academic integrity and collaboration

Laboratory teams share procedures and raw measurements; analysis, figures, and submitted reports are individual unless an assignment says otherwise. Copying another student's calculation, figure, notebook entry, or report language is unauthorized collaboration.

Accessibility, health, and support

Access and Inclusion Services coordinates academic accommodations. Students with allergies, pregnancy-related considerations, mobility needs, or other laboratory access concerns should contact the instructor privately so an appropriate safety plan can be established. Counseling, tutoring, food support, and emergency grants are available through the Student Care Network.

Weekly course schedule

Complete assigned preparation before the first meeting listed for the week. Due items are repeated here as a planning aid; the assignment calendar remains the authoritative list.

Week	Dates	Focus and preparation	Due this week
1	Aug 24-30	Cell theory; scale and units; microscopy orientation; safety contract. Read Chs. 1-2.	Problem Set 1 (Sun)
2	Aug 31-Sep 6	Chemical bonds, water, and macromolecular structure; pipetting and dilution practice.	Notebook setup (Wed)
3	Sep 7-13	Protein structure and molecular recognition; microscopy image calibration.	Problem Set 2 (Sun)
4	Sep 14-20	Membrane architecture, diffusion, osmosis, and electrochemical gradients.	Report 1 analysis workshop (Wed)
5	Sep 21-27	Transport proteins and tonicity; red-cell and plant-cell observations.	Lab Report 1 (Wed); Problem Set 3 (Sun)
6	Sep 28-Oct 4	Organelles and intracellular trafficking; interpreting localization evidence.	Notebook Check 1 (Wed)
7	Oct 5-11	Unit synthesis and assessment; introduction to energy and catalysis.	Unit Exam 1 (Wed)
8	Oct 12-18	Enzyme kinetics, inhibition, temperature, and pH; spectrophotometry.	Problem Set 4 (Sun)
9	Oct 19-25	ATP coupling, redox reactions, and cellular respiration.	Enzyme report figure check (Wed)
10	Oct 26-Nov 1	Glycolysis, citric acid cycle, and oxidative phosphorylation.	Lab Report 2 (Wed); Problem Set 5 (Sun)
11	Nov 2-8	Photosynthesis; comparing chemiosmosis across organelles.	Notebook Check 2 (Wed)
12	Nov 9-15	DNA organization, replication, and repair; gel interpretation.	Problem Set 6 (Sun)
13	Nov 16-22	Transcription, translation, and regulation; unit assessment.	Unit Exam 2 (Wed)
14	Nov 23-29	No Monday lecture; asynchronous gene-expression case; no Wednesday lab.	Problem Set 7 (Sun)
15	Nov 30-Dec 6	Cell signaling, receptors, and feedback; membrane-transport investigation.	Notebook Check 3 (Wed); Problem Set 8 (Sun)
16	Dec 7-11	Cell cycle, checkpoints, cancer, and course synthesis.	Lab Report 3 (Wed)
Finals	Dec 14-18	Cumulative final examination period.	Final Exam (Wed, Dec 16)

End-of-term reminder

Plan backward from the final due date, use office hours before a deadline becomes urgent, and contact the instructor early when circumstances interfere with your work. Asking for clarification is part of successful course participation.